

The technical and economic feasibility of an off-grid solar PV storage system to power a refugee camp near Damascus the capital of Syria



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Introduction

Given its location, which benefits from lots of sunshine, Syria realizes that solar power is a smart way to meet its growing energy needs. Using solar energy also helps protect the environment and reduces Syria's need for fossil fuels. The daily average sunshine, or solar radiation, in the Damascus area is very good, sitting around 5 to 5.5 kWh/m² per day, whereas the diffuse solar irradiance ranges from 1.45 to 1.55 kWh/m² per day [1].

Syria has great solar resources, making large solar farms and small, independent systems—like the one planned for the refugee camp—possible. Solar power isn't just a practical way to make electricity; it also supports Syria's environmental goals. By using less fossil fuel, solar energy makes Syria cleaner and more sustainable, reducing the harm caused by traditional power plants.

The Syrian Refugee Camp near Damascus is home to about 100,000 people. This community needs a consistent and reliable power source for all its basic services, including water, healthcare, and education. To meet these daily energy needs, a dependable and sustainable electricity supply is absolutely necessary. Energy Solutions consulting group has been hired to study the technical and financial feasibility of an off-grid solar PV system with storage for the camp near Damascus. The main goal of this project is to find the most cost-effective and clean power technologies for this challenging location.

will explain how we sized the solar panels and the battery storage system. Finally, we will compare the solar system's costs against older power sources, like diesel generators, to give you a final, clear recommendation.

Assumptions

Five areas of the Syrian refugee camp near Damascus use energy: 4 clinics, 3 water pumping facilities, 5 schools, 100 street lighting units, and prefabricated structures that can house 100,000 people. The next subsections address the general presumptions and energy requirements for each part.

According to the problem statement, there were 100,000 people living in the camp in the beginning. Since most people do not live in the camp for an extended period of time, it is crucial to remember that this number is regarded as a static capacity. Over time, the population's constant influx and outflow of individuals successfully balances each other out. The camp is open seven days a week, twenty-four hours a day.

Energy requirement of refugee prefab units

One television set with a rated power of 50 watts is used in the prefabricated apartments designed for five people. A research that looked at average daily TV viewing hours was taken into consideration in order to calculate power usage and found that people watch TV for an average of 4 hours per day. As seen in Table 1, this four-hour period was considered while calculating energy.

The climate patterns in Damascus, Syria, were analyzed to determine the operational hours for fans and lighting systems. Due to the proximity and similar semi-arid climate of the region, the usage hours for fans and LED lights calculated from existing regional data (9 hours/day for fans and 6.5 hours/day for LEDs) are assumed to be consistent for the camp near Damascus. According to this data, the average temperatures throughout a six-month period from May to October are higher than what is deemed

acceptable for people. As a result, the fan must run constantly throughout this period, with an approximate daily usage of 18 hours. The fan's projected average annual running time is nine hours per day, given that it runs for eighteen hours every day for six months of the year and is not required for the remaining six months.

To maintain its temperature, a normal residential refrigerator runs for around eight hours every day. According to this, refrigerators typically operate for eight hours per day, based on a duty cycle of 35%. The prefabricated units are designed to house 100,000 people, with each unit accommodating 5 individuals. Consequently, a total of 20,000 prefabricated units are needed. It is crucial to consider this quantity when assessing the energy requirements for each prefabricated unit. The outcomes of the energy consumption for these units are presented in Table 1.

Table 1. Daily energy consumption for 100,000 people						
Load type	Units	Rated Power (W/unit)	Usage hours per day	Energy kWh/day per prefab	Number of prefab units	Total energy (kWh/day)
TV	1	50	4	0.2	20000	4000
Fan	2	50	9	0.9	20000	18000
LED	5	10	6.5	0.325	20000	6500
Small Refrigerator	1	200	8	1.6	20000	32000
						60500

Energy requirements of clinics

The Syrian refugee camp near Damascus has a total of four clinics. Refugees may get primary healthcare, birthing services, and round-the-clock emergency medical help at the camp's medical facilities. Because it holds medications, vaccinations, and other temperature-sensitive products, the hospital/clinic refrigerator works at a 40% duty cycle for around nine hours per day. It is expected that LED lights will run constantly, 24 hours a day, as the clinic is open 24/7 to service emergency patients. Six hours a day are dedicated to the operation of medical equipment. Defibrillators, ventilators, and imaging equipment are examples of medical equipment that is not always in use. The total energy consumed by all clinics is 615.6 kWh/day.

Table 2. Daily energy consumption for 4 clinics operating 24/7						
Load type	Units	Rated Power (W/unit)	Usage hours per day	Energy kWh/day per clinic	Number of clinics	Total energy (kWh/day)
Refrigerator	2	750	9	13.5	4	54
LED Lights	10	10	24	2.4	4	9.6
Medical Equipment	2	10500	6	126	4	504
Others	1	500	24	12	4	48
						615.6

Energy requirements for pumping water supply

Pumping water from boreholes provides daily water consumption. It is believed that a pump operates for around eight hours every day on average. Since the control panels for water supply pumps are in charge of keeping an eye on and managing the pumps' functioning, they are usually made to run continuously or around-the-clock. In order to maintain water pressure and satisfy demand, these control panels are crucial

for controlling the water supply since they make sure that pumps start and stop as necessary. Even though the pumps themselves might only run eight hours a day, the control panels that oversee the complete water delivery system are always in operation. The energy usage for three water pumping stations is shown in Table 3. The sum of the three pumping stations' daily energy use is 2520 kWh/day.

Table 3. Daily energy consumption for water pumping stations						
Load type	Units	Rated Power (W/unit)	Usage hours per day	Energy kWh/day for pumping station	Number of pumping stations	Total energy (kWh/day)
Electric Motors	3	25000	8	600	3	1800
Control Panel	2	5000	24	240	3	720
						2520

Energy requirements for schools

The camp near Damascus currently has five schools. Schools must work two shifts, each lasting five hours, to meet student educational demands. To provide students with adequate illumination for studying, the school requires LED lights for ten hours every day. An estimated 4 hours a day, 2 hours every school shift, are spent in the other designated areas. The energy usage for 5 schools is 750 kWh per day, as shown in Table 4.

Table 4. Daily energy consumption for 5 schools						
Load type	Units	Rated Power (W/unit)	Usage hours per day	Energy kWh/day per school	Number of schools	Total energy (kWh/day)
LED Lights	500	10	10	50	5	250
Science Lab	50	500	4	100	5	500
						750

Energy requirements for street lighting

The streetlights should be on for an average of 11 hours, based on an estimate of the typical dawn and sunset hours. An approximate lighting schedule of 12 hours for 6 months of the year and 10 hours for the remaining 6 months is assumed for the Damascus region. Table 5 shows that 165 kWh of energy is used daily for street lighting.

Table 5. Daily energy consumption for street lighting					
Load type	Units	Rated Power (W/unit)	Usage hours per day	Energy kWh/day	Total energy (kWh/day)
LED Lights	100	150	11	165	165

Total energy requirements

The energy use is **64550.6 kWh per day**, or **23.59 GWh annually**, and **1.966 GWh per month**. The daily energy need would serve as the foundation for the solar panel design calculations in the parts that follow. It is important to remember that this energy need is the quantity of AC energy that must be supplied, while the solar panels supply DC energy. The percentage of energy used in each area of the camp for refugees shows that **93.72%** of energy consumption is attributed to prefabricated units, meaning that the majority of energy must be provided to these units. The remaining **6.28%** of camp energy consumption is attributed to schools, pumping stations, clinics, and streetlights. Making well-informed decisions during the solar panel design process is facilitated by this study. It ensures that the solar power system is not only

appropriately sized but also strategically allocated to meet the predominated energy needs, particularly within the prefabricated units.

Number of required solar panels

Damascus, Syria has a latitude of approximately 33.5° N 36.3° E. The optimal tilt angle for maximum annual energy production is generally set close to the site's latitude, so it is assumed to be 33.5° from horizontal. The panels are fixed at this angle (without tracking) and are facing south (Azimuth 180°) to ensure the highest solar absorption. The solar cell type assumed is crystalline silicon, and the panels are 315 W units with negligible annual degradation.

The region's average daily peak sun hours for each month of the year are shown in Table 6, along with the quantity of panels needed to meet the camp's energy needs. The daily energy use must first be converted from AC 64,550.6 kWh/day to DC, assuming an inverter efficiency of 97%. The largest number of panels, 42,260 panels, was collected during the month with the lowest solar peak hours (December). Notably, the cost estimates will utilize the 1 kWp produced by four 315 W panels.

Month	Peak sun hours daily	number of days	kWh/day (AC)	kWh/month (DC)	Required Panel Power (kWp)	Number of 315 W panels
Jan	5.1	31	64,550.6	2,062,957	13,015.6	41,365
Feb	5.8	28	64,550.6	1,863,316	11,440	36,367
Mar	7.7	31	64,550.6	2,062,957	8,639.8	27,458
Apr	9.4	30	64,550.6	1,996,410	7,086.6	22,506
May	9.8	31	64,550.6	2,062,957	6,757.5	21,475
Jun	11.8	30	64,550.6	1,996,410	5,639	17,911
Jul	12	31	64,550.6	2,062,957	5,545.6	17,615
Aug	11.7	31	64,550.6	2,062,957	5,699.4	18,107
Sep	10	30	64,550.6	1,996,410	6,654.7	21,139
Oct	8.8	31	64,550.6	2,062,957	7,596.1	24,142
Nov	6.2	30	64,550.6	1,996,410	10,733.4	34,115
Dec	5	31	64,550.6	2,062,957	13,309.4	42,260

Number of areas for the solar panels

One 60 cell 315 W solar panel requires 1.7 m² of space in total. All 42260 panels require a total space of 71842 m², or 17.75 acres. One football field has a size of 1.9 acres, therefore there would be around 9.34 football fields in all.

Daytime and Nighttime energy requirements

The camp is open twenty-four hours a day. The energy required for a 24-hour period is supplied by solar panels. Lithium-ion batteries must store a portion of the energy to supply the electricity required at night. In order to decide how much should be allocated to nighttime usage and how much to daytime usage, the plant's energy consumption was reviewed to account for the energy used throughout the night.

The energy needed for the prefabricated units during the day and at night are shown in Table 7. It is expected that people watch TV for two hours during the day and two hours at night. It is believed that the

fan runs for four hours at night and five hours during the day. The refrigerator is used for half of the day and half of the night, while the lights are utilized for 5.5 hours at night and 1 hour during the day. 31,500 kWh/day is the total amount of energy needed at night when batteries are needed to supply power to all prefabricated units.

Table 7. Prefab Units Nighttime & Daytime Energy						
Load type	Total daily energy (kWh/day)	Night usage (hour)	Total usage (hours)	Night energy ratio	Night energy (kWh/day)	Day energy (kWh/day)
TV	4000	2	4	0.5	2,000	2,000
Fan	18000	4	9	0.4	8,000	10,000
Small Refrigerator	32000	4	8	0.5	16,000	16,000
LED Lights	6500	5.5	6.5	0.8	5,500	1,000
					31,500	29,000

The energy requirements for clinics throughout the day and at night are shown in Table 8. Clinics use a total of 55.8 kWh of electricity per day for night operations.

Table 8. Clinics Nighttime & Daytime Energy						
Load type	Total daily energy (kWh/day)	Night Usage (h)	Total Usage (h)	Night Energy Ratio	Night Energy (kWh/day)	Day Energy (kWh/day)
Refrigerator	54	4.5	9	0.5	27	27
LED Lights	9.6	12	24	0.5	4.8	4.8
Medical Equipment	504	0	6	0	0	504
Others	48	12	24	0.5	24	24
					55.8	560.8

The energy consumption of three pumping stations during the day and at night is shown in Table 9. Pumping stations that require battery power have an energy requirement of 360 kWh per day.

Table 9. Pumping Stations Nighttime Energy						
Load type	Total daily energy (kWh/day)	Night Usage (h)	Total Usage (h)	Night Energy Ratio	Night Energy (kWh/day)	Day Energy (kWh/day)
Electric Motors	1800	0	8	0	0	1800
Control Panel	720	12	24	0.5	360	360
					360	2160

The total daily energy consumption for the five Schools is 750.0 kWh/day (from Table 4). Since the text specifies that the energy is supplied during the day to align with school shifts, the entire load is a Day Energy demand.

Table 10. Schools Day and Night Energy						
Load type	Total Daily Energy (kWh/day)	Night Usage (h)	Total Usage (h)	Night Energy Ratio	Night Energy (kWh/day)	Day Energy (kWh/day)
LED Lights	250	0	10	0	0	250
Others	500	0	4	0	0	500
					0	750

The total daily energy consumption for Street Lighting is 165.0 kWh/day (from Table 5). Since street lighting operates entirely when the sun is down, the entire load is a Night Energy demand

Table 11. Street Lighting Day and Night Energy						
Load type	Total Daily Energy (kWh/day)	Night Usage (h)	Total Usage (h)	Night Energy Ratio	Night Energy (kWh/day)	Day Energy (kWh/day)
LED Lights	165	11	11	1	165	0
					165	0

The off-grid solar PV system requires a large battery bank to supply power during nighttime hours 32,080.8 kWh/day and periods of low solar radiation. For this large-scale project, industrial deep-cycle lithium-ion cells are assumed due to their longevity and capacity. The final battery sizing relies on the following key design parameters established in the original study:

Total DC Energy Required: 66,547 kWh/day

System Voltage: 480 V

Days of Autonomy (DA): 2 days

Depth of Discharge (DOD): 70% (or 0.7)

Battery Efficiency: 90% (or 0.9)

Assumed Battery Unit: 2 V with 2000 Ah capacity

$$\text{Total Capacity (Ah)} = \frac{66547 \text{ Wh/day} \times 2 \text{ days}}{480 \text{ V} \times 0.7 \times 0.9} = 440.12 \text{ Ah}$$

This capacity requires:

- **240** batteries connected in series to achieve the 480 V system voltage.
- **221** parallel strings (to meet the required Ah capacity).

So the system requires a total of $240 \times 221 = 53040$ batteries for two days of continuous operation for the camp.

Diesel Generator

It is predicted that generators last between 20 and 25 years. A \$2000~\text{kW} diesel generator can consistently provide 1000 kW of electricity while running at half load. The fuel consumption is \$273.3\$ gallons per hour, which is approximately 1034.5 L/hr (1 gallon=3.785 L). This specific fuel consumption is about 1.0345L/kWh when operating at 1000 kW output.

A total of four actively functioning generators, providing a total capacity of 4000 kW (or 96,000 kWh/day), and an extra four standby generators are required to maintain an uninterrupted power supply to meet the camp's continuous energy requirements. This redundancy is vital for handling system malfunctions or maintenance outages, guaranteeing continuous power for the camp's 64,550.6 kWh/day energy demand.

Based on the recalculated energy requirement of 64,550.6 kWh/day, the required fuel volume over the project's lifetime is significant. When scaled to meet the exact load requirement, the total amount of diesel required for a span of 25 years is calculated to be 603.2 million L.

Table 12. Number of generator and diesel required for 25 years									
Generator size (kW)	delivered power (kW)	Fuel (L)	Daily energy (kWh/day)	Energy required kWh/day	number of generators	Diesel per day (L)	Diesel for 4 generators in 1 day (L)	Diesel for 4 generators for 1 year (million L)	Diesel for 4 generators for 25 years (million L)
2000	1000	1034.5	24000	64550.6	4	6663.8	24828	66762	603.2

Balance of Plant (BoP)

All of the elements necessary for the system to function are included in the balance of the plant. These structures for solar PV systems include mounting systems that secure the panels, wiring that links the panels to storage or inverters, inverters that transform the DC power from the panels into AC electricity that can be used in residences, and monitoring and control systems.

Any power production project must include the balance of plants because it guarantees the safe and effective delivery of the electricity produced by solar panels. In the economic analysis portion, the cost analysis of the remaining factories is finished.

Environmental impact assessment (diesel electric motors & solar panels)

The pollutants released through electricity generation from diesel and the amounts are shown in Table 13. For 64,550.6 kWh/day, the emissions are calculated daily and for the span of 25 years. This lifetime analysis reveals the substantial environmental cost of relying on diesel power generation as compared to the solar PV alternative.

Table 13. Pollutant and emission amount over the span of 25 years of camp's operation								
Pollutant	CO ₂	CO	NO _x	SO ₂	VOC	PM ₁₀	CH ₄	N ₂ O
Emission g/kWh	690	5	9.8	0.007	0.27	0.26	0.02	0.09
Emission kg/day	44,535.4	322.5	632.6	0.45	17.43	16.78	1.29	5.81
Emission ton/year	16,255.4	117.8	230.9	0.16	6.36	6.12	0.47	2.12
Emission tons in 25 years	406,385.6	2,946	5,772.3	4.1	159.1	153.1	11.8	53.1

When solar panels are operating normally, they don't emit any pollutants. They produce electricity by using a mechanism called the photovoltaic (PV) effect, which turns sunlight into electrical energy. Since no fuels are burned during this clean process, solar panels are an eco-friendly way to generate power.

However, during the course of the panels' lifetime, a very little quantity of CO₂ is released. During the first few years of operation, 42 g CO₂/kWh generated. Even if solar panel manufacture is taken into account

and the emissions are released over a five-year period, the total would still be far less than the emissions from diesel generators, which resulted in over 406,000 tons of CO₂ over 25 years.

Economic analysis: Capital Costs

solar system

The new total installed PV capacity is based on the 64,550.6 kWh/day energy requirement and the lowest daily Peak Sun Hours (assumed 5.0 h/day for the winter months), resulting in a net DC requirement of 13,310 kWp. The problem specifies the PV panel price is \$300 kWp.

A solar system costs \$300 per kWp. It was determined that a maximum of 42,254 panels will be needed to supply the energy needs of the camp near Damascus. The total required capacity is 13,310 kWp.

$$\text{Total PV Cost} = 13,310 \text{ kWp} \times \frac{\$300}{\text{kWp}} = \$3,993,000$$

Solar panels will set you back **\$3.993 million** in initial investment.

Batteries

The calculation is based on the new night-time energy requirement of 32,476.5 kWh/day and the specified price of \$200 kWhr for Li-ion batteries with a 13.5 kWh capacity (10.8 kWh usable energy at 80% DoD). The system requires 3,007 batteries, with 2.5 replacement cycles over 25 years.

A total of 3,007 batteries were determined. The lifespan of a lithium-ion battery is 10 years, and each battery costs \$200 kWhr. Over a 25-year period, batteries will cost \$20.3 million in total.

Table 14. Cost of batteries for storage over a span of 25 years				
Battery Capacity (kWh)	Nighttime energy requirements (kWh/day)	Number of batteries	Cost of one year (million \$)	Cost for 25 years (million \$)
13.5	32,476.5	3007	0.812	20.3

Inverter, mounting, labor, wiring

The BoP costs are based on the total installed PV capacity, which is 13,310 kWp (13,310,000 W) for the new system. The original \$/W factors are maintained as reasonable assumptions for BoP components in the Middle East region.

$$\text{Power} = 13,310 \text{ kWp} \times 1000 = 13,310,000 \text{ W}$$

Table 15. Cost of inverter, wiring, labor, and mounting			
	Cost (\$/W)	Power (W)	Total cost (\$ million)
Inverter	0.18	13,310,000	2.396
Wiring	0.1	13,310,000	1.331
Labor	0.5	13,310,000	6.655
Mounting	0.07	13,310,000	0.932
			11.314

Total cost of solar system

Wiring, labor, mounting, solar panel installation, storage, and inverter costs are all included in the overall cost of the solar system.

The total initial cost of the solar system is the sum of the initial PV panel cost (\$3.993 million), the BoP cost (\$11.314 million), and the initial battery cost (\$8.12 million). So the entire initial amount spent came to \$23.427 million.

As can be seen in table 16, Most of the solar system's cost includes labor, mounting, inverter, cables, panels, and initial battery bank.

Table 16. Total Cost				
Component	PV Panels	BoP (Inverter, Labor, etc.)	Batteries	Total
Cost	\$3.993 million	\$11.314 million	\$ 8.12 million	\$23.427 million

Diesel electric generators

It was determined that 603. 2 million L of diesel would be used during a 25-year period to meet the camp's energy needs. Diesel fuel near Damascus is estimated to cost \$0.801 per L, accounting for current market conditions in Syria.

$$\text{Total Diesel Fuel Cost (25 years)} = 603.2 \times 10^6 \text{L} \times \$0.801 / \text{L} = \$483.16 \text{ million}$$

The cost of each 2000 kW generator is around \$300,000 and the total cost of the eight required generators (four active, four standby) would be approximately \$2.4 million.

So the total lifetime cost of the diesel system, including fuel and capital expenditure, is \$485.57 million.

Grid electricity generation

The Syrian residential power tariff is extremely low due to government subsidies, estimated at \$0.019 per kWh. This price is used for comparison as the project is a humanitarian effort.

$$\text{Daily Cost} = 64,550.6 \text{ kWh/day} \times \$0.019 \text{ kWh} = \$1,226.46$$

$$\text{Annual Cost} = \$1,226.46 / \text{day} \times 365 \text{ days} = \$447,657.9 \rightarrow \text{over 25 years} = \$11.19 \text{ million}$$

Table 17. Cost of on grid electricity				
Cost of electricity \$/kWh	Energy kWh/day	Cost of electricity per day	Cost of electricity per year	Cost of electricity in 25 years
\$0.019	64,550.6	\$1,226.46	\$447,657.9	\$11.19 million

Cost Comparison

The comparison of the three energy generation scenarios demonstrates a clear economic advantage for both the off-grid solar PV system and the heavily subsidized on-grid option. When analyzing the total cost of ownership over the 25-year project lifespan, the diesel generator system is overwhelmingly expensive due to high fuel consumption and maintenance.

The 25-year cost for the solar PV system (\$35.61 million) is significantly lower than the diesel generator option (\$485.57 million). However, the subsidized Syrian grid electricity, at a total of \$11.2 million, presents the lowest energy cost, but this figure excludes the enormous infrastructure CAPEX required to connect the remote camp, which would push its true cost higher. The solar system amortizes its high initial CAPEX via zero fuel costs, resulting in a lifetime Levelized Cost of Electricity of \$0.076 kWh.

Final Conclusion

The techno-economic assessment for providing energy to the 100,000 refugees near Damascus clearly supports the implementation of the off-grid solar PV system with energy storage.

Key Findings:

Cost Efficiency: The total lifetime cost (25 years) of the solar PV system is \$35.61 million. This is overwhelmingly more cost-effective than the alternative diesel generator system, which totals \$485.57 million due to massive ongoing fuel costs.

Environmental Impact: The diesel alternative would emit over 406,000 tons of CO₂ over 25 years. The solar PV system operates with near-zero emissions, providing a sustainable solution in line with UN environmental goals.

Grid Constraint: While the heavily subsidized Syrian grid tariff (\$11.2 million lifetime cost) is nominally the cheapest, this figure excludes the enormous capital cost and feasibility challenges of connecting a remote camp to the country's damaged national infrastructure.

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